



Modelling Financial Markets

Chapter 13: Machine Learning in Finance



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/3)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CPCV: Combinatorial Purged Cross-Validation
- ▣ CRSP: Center for Research in Security Prices
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DML: Double/Debiased Machine Learning
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ EV: Explained Variation
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FFD: Fixed-width window Fractional Differentiation

Acronyms used in this material (2/3)

- ▣ **FFN: FeedForward Network**
- ▣ **FN: False Negative**
- ▣ **FRED-MD: Federal Reserve Economic Data, Monthly Database (McCracken–Ng)**
- ▣ **FWER: Family-Wise Error Rate**
- ▣ **GAN: Generative Adversarial Network**
- ▣ **GBM: Gradient Boosting Machine**
- ▣ **GRS: Gibbons–Ross–Shanken (test)**
- ▣ **GRU: Gated Recurrent Unit**
- ▣ **GW: Giacomini–White (test)**
- ▣ **HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)**
- ▣ **HML: High Minus Low (value factor)**
- ▣ **IBM: International Business Machines (Corporation)**
- ▣ **IS: In-Sample**
- ▣ **LASSO: Least Absolute Shrinkage and Selection Operator**
- ▣ **LIME: Local Interpretable Model-agnostic Explanations**
- ▣ **LLM: Large Language Model**
- ▣ **LS: Linear SDF estimated by unregularised Least Squares**
- ▣ **LSTM: Long Short-Term Memory (network)**
- ▣ **MA: Moving Average**
- ▣ **MDA: Mean Decrease Accuracy**
- ▣ **MDI: Mean Decrease Impurity**
- ▣ **ML: Machine Learning**
- ▣ **MLP: MultiLayer Perceptron**
- ▣ **MSCA: Marie Skłodowska-Curie Actions**
- ▣ **OLS: Ordinary Least Squares**
- ▣ **OOB: Out-Of-Bag (error)**

Acronyms used in this material (3/3)

- OOS: Out-Of-Sample
- PAC: Probably Approximately Correct (learning)
- PBO: Probability of Backtest Overfitting
- PCR: Principal Component Regression
- PLS: Partial Least Squares
- PSR: Probabilistic Sharpe Ratio
- PT: Pesaran–Timmermann (test of directional accuracy)
- RC: Reality Check (White's test for data snooping)
- RF: Random Forest
- RL: Reinforcement Learning
- RMW: Robust Minus Weak (profitability factor)
- ROC: Receiver Operating Characteristic (curve)
- RSI: Relative Strength Index
- SD: Standard Deviation
- SDF: Stochastic Discount Factor
- SE: Standard Error
- SHAP: SHapley Additive exPlanations
- SPA: Superior Predictive Ability (test)
- SR: Sharpe Ratio
- TN: True Negative
- TP: True Positive
- US: United States
- VIX: CBOE Volatility Index
- WF: Walk-Forward (validation)
- XAI: eXplainable Artificial Intelligence

Outline

Motivation

Machine Learning Toolbox: A Recap

Measuring and Testing Predictive Ability

Financial Data and Features

Labelling

Cross-Validation in Finance

Case Study: S&P 500

Backtest Overfitting

Machine Learning for Inference

Frontiers

Case Study: Deep Learning in Asset Pricing

AI for Scientific Discovery

Best Practices and Conclusions

Why Machine Learning in Finance?

The promise of machine learning (ML)

- ▣ Flexible, non-linear functional forms
- ▣ Hundreds of candidate predictors (the “factor zoo”, Chapter 3)
- ▣ Interactions discovered automatically
- ▣ Alternative data: text, images, order books, on-chain data

The problem

- ▣ Very low **signal-to-noise ratio**
- ▣ **Non-stationarity**: regimes, structural breaks
- ▣ Markets **adapt**: signals decay once exploited
- ▣ Few independent observations
- ▣ Massive risk of **overfitting** (fitting noise)

Key message of this chapter

- ▣ In finance, the hard part is not fitting a model — it is **not fooling yourself**: correct labels, correct validation, honest backtests

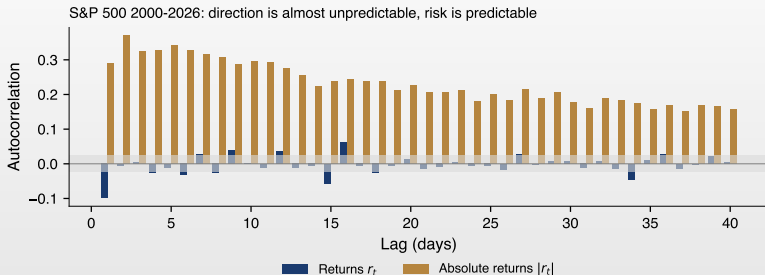
Finance Is Not Image Recognition

	Typical ML domain	Financial markets
Signal-to-noise	high (a cat is a cat)	extremely low
Data-generating process	stable	changes over time, reacts to users
Sample size	millions of i.i.d. examples	one history, serially dependent
Labels	given by humans	must be constructed
Cost of error	misclassification	real money, leverage, ruin
Experimentation	new data can be generated	cannot re-run history

Adaptive Markets Hypothesis: Lo (2004)

- Predictability is not a constant: it appears, is arbitrated away, and re-appears under new conditions
- A model that worked yesterday is a **hypothesis**, not a law (market efficiency: Chapter 2)

What Is Predictable? Direction vs. Risk



- Autocorrelation at lag k : $\rho_k = \text{Corr}(x_t, x_{t-k})$; the grey band is the 95% no-correlation band $\pm 1.96/\sqrt{T}$
- Autocorrelations of returns r_t are tiny (inside the band); those of $|r_t|$ are large and persistent
- ML has the best chance where there is structure: **volatility, risk, liquidity, cross-sectional ranking**

Evidence from Empirical Asset Pricing

Gu, Kelly & Xiu (2020), *Review of Financial Studies*

- Predict **monthly returns of individual American stocks**; compared: linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS), random forests, boosted trees, neural networks
- Out-of-sample $R_{OOS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$ is small (well below 1% per month)
- Trees and neural nets beat linear models; gains come from **non-linear interactions**; dominant predictors: momentum, liquidity, volatility
- A small R_{OOS}^2 can still rank thousands of stocks well enough for large portfolio gains (next slide)

Caveat

- For a **single time series** (one index) the effective sample is far smaller and the evidence much weaker — as the S&P 500 case study shows

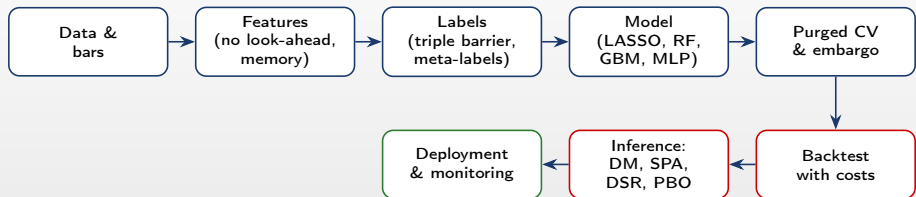
How Small Is Small? R^2_{OOS} and Economic Value

- **Campbell & Thompson (2008)**: in the model $r_{t+1} = \mu + x_t + \varepsilon_{t+1}$, with $R^2 = \text{Var}(x_t)/\text{Var}(r_{t+1})$ and constant $\text{Var}(\varepsilon_{t+1})$, an unconstrained mean-variance investor who knows x_t (no costs, no estimation error) reaches

$$SR^{*2} = \frac{SR_0^2 + R^2}{1 - R^2}, \quad SR_0 = \text{Sharpe ratio of buy-and-hold}$$

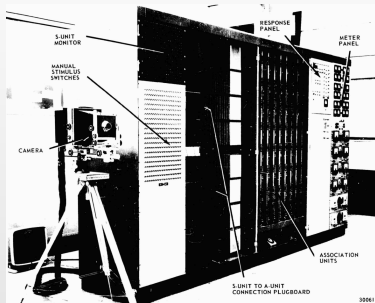
- S&P 500, monthly log returns 2001–2026: $SR_0 = 0.128$ per month (0.44 annualised)
 - ▶ idealised illustration: $R^2 = 0.5\%$ per month $\Rightarrow SR^* = 0.51$ annualised (+15%); $R^2 = 1\% \Rightarrow 0.57$ (+27%)
- The benchmark matters: **Welch & Goyal (2008)** show that most published predictors lose to the **historical mean** out of sample
 - ▶ **Gu, Kelly & Xiu (2020)** measure R^2_{OOS} against a zero forecast, as does Chapter 14; the historical mean is a harder benchmark
- A monthly R^2 of 0.5% is economically large, but sampling noise is of the same order: for the *in-sample* OLS R^2 of one predictor with no skill (intercept, independent errors, $n \approx 300$ months), $nR^2 \approx \chi_1^2$, so R^2 has mean $1/n \approx 0.3\%$ and SD $\sqrt{2}/n \approx 0.5\%$; R^2_{OOS} can be negative, has no such null distribution and needs its own test (Clark–West)

The Financial ML Workflow



- Sections 4–6 follow the top row, Sections 7–8 the bottom row; Section 3 gives the tests used throughout
- **Look-ahead bias**: using information not available at decision time; **selection bias**: reporting the best of many trials
- López de Prado (2018): most ML failures in finance are due to the process, not to the algorithm

From the Perceptron to Deep Blue



Mark I Perceptron, operator's manual (1960)

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons



IBM Deep Blue, Computer History Museum

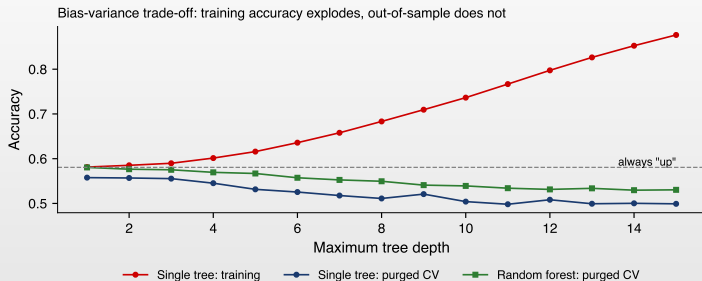
Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons

- **Perceptron**: a learning rule for a single artificial neuron [Rosenblatt \(1958\)](#); the Mark I was hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov [Campbell, Hoane & Hsu \(2002\)](#)
- Deep Blue searched with hand-built rules; modern ML (this chapter) *learns* the rules from data

Prerequisites Recap: Risk Minimisation, Bias–Variance, CV

- ▣ Supervised learning: $\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_i L(y_i, f(x_i)) + \lambda \Omega(f)$; loss L (squared error, log-loss), complexity penalty Ω , tuning $\lambda \geq 0$
- ▣ Expected error = bias² + variance + irreducible noise σ^2 (Hastie, Tibshirani & Friedman (2009), Ch. 7)
 - ▶ In finance σ^2 dominates: a small bias reduction is rarely worth a large variance increase
- ▣ In-sample (IS) error is optimistic; out-of-sample (OOS) error is what matters
- ▣ K -fold cross-validation (CV) chooses λ and compares models; a final test set is never touched
 - ▶ Assumes exchangeable observations: Section 6 shows when this holds for time series and when it fails
- ▣ This chapter: classification of the 5-day direction; logistic regression (logit) $p = 1/(1 + e^{-x^\top \beta})$ is the linear benchmark

Bias–Variance in Practice: Tree Depth



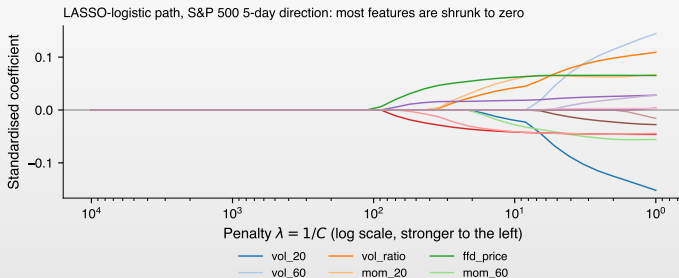
- S&P 500, 5-day direction; accuracy = share of correctly classified days
- Training accuracy rises to $\approx 87\%$ with depth; purged-CV accuracy **falls** towards 50%
- Random forests degrade more slowly, but never beat the “always up” baseline (dashed line)

Recap: Penalised Regression, Trees, Forests, Boosting, Nets

- Penalised least squares / logit: $\lambda [\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2]$
 - ▶ $\alpha = 0$ Ridge (Hoerl & Kennard (1970)), $\alpha = 1$ LASSO (Tibshirani (1996)), in between Elastic Net (Zou & Hastie (2005)); standardise features on the training folds only
- Trees: recursive binary splits minimising Gini impurity; low bias, high variance
- Random forest (RF): average of B decorrelated bootstrap trees (Breiman (2001))
 - ▶ Out-of-bag (OOB) error is **optimistic** when labels overlap: bootstrap draws near-duplicates
- Gradient boosting machine (GBM): $F_m = F_{m-1} + \eta h_m$, small trees fitted to the negative gradient (Friedman (2001))
- Multilayer perceptron (MLP): $h^{(\ell)} = g(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$, ReLU g , early stopping; shallow nets suffice on tabular returns (Gu, Kelly & Xiu (2020)); architectures in Chapter 14
- What is new for this audience: **LASSO selection is not inference** (Section 9), and all CV scores need a test (Section 3)

 Random Forests

LASSO Regularisation Path



- L_1 -penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Strong penalty: every coefficient is zero; the first feature to enter is the FFD price, a scaled price level (Section 4)
- Weak penalty: the volatility features (vol_20, vol_60, vol_ratio) get the largest coefficients
- All coefficients are small in absolute value: weak signal, consistent with near-efficient markets

Worked Example: How a Forest Averages Its Trees

- Random forest of the case study (400 trees, minimum leaf size 100), trained on the first four purged folds
- Prediction for the last observation (11 September 2026): $P(\text{up in 5 days})$ from each tree
 - ▶ individual trees range from 0.31 to 0.88 (standard deviation 0.11)
 - ▶ 81.5% of the trees vote “up” ($p > 0.5$)
 - ▶ forest probability = average of the 400 trees = 0.608
- Single trees disagree strongly; the average is far more stable
- Stability is not accuracy: the forest still does not beat “always up” (Section 7)

Recap: Machine Learning Toolbox

- ▣ Supervised learning = loss + complexity penalty, tuned by cross-validation
- ▣ Finance is noise-dominated: variance control matters more than flexibility
- ▣ Toolbox: penalised logit (Ridge / LASSO / Elastic Net), trees, random forests, gradient boosting, MLP
- ▣ A CV score is an **estimate**: it has a sampling distribution
- ▣ Next: how to **test** whether a classifier beats a benchmark

Worked Example: The Model That Learned “Always Up”

	Pred. up	Pred. down
Actual up	TP = 3,572	FN = 68
Actual down	FP = 2,591	TN = 38

- L2-penalised logit, S&P 500 case study (Section 7), out-of-fold predictions of purged 5-fold CV, $n = 6,269$
- TP, FN, FP, TN: true positives, false negatives, false positives, true negatives

- Accuracy = $(TP + TN)/n = 57.6\%$
- Precision = $TP/(TP + FP) = 0.580$; recall = $TP/(TP + FN) = 0.981$; $F_1 = 0.729$
- The model predicts “up” on 98.3% of days: it has essentially learned “always up”
- Is 57.6% evidence of skill? Two questions:
 - ▶ compared with what (next slide)?
 - ▶ with which sampling error (the tests that follow)?

Always Compare with a Naive Baseline

- ▣ “**Always up**” baseline: predict the majority class every day
- ▣ Its accuracy equals the share of up-labels: $\bar{y} = \frac{1}{n} \sum_i y_i$
- ▣ S&P 500, 5-day direction, 2001–2026: $\bar{y} = 58.1\%$ — a “57%-accurate” model is *worse* than doing nothing
- ▣ Other baselines: yesterday’s direction, a moving-average rule, buy-and-hold for trading
- ▣ A model has value only if it beats the baseline **out of sample**, after costs, by more than sampling error

Testing Directional Accuracy: Pesaran–Timmermann

□ H_0 : the predicted sign x_t is independent of the realised sign y_t (Pesaran & Timmermann (1992))

□ $\hat{P}$ = share of correct signs; under H_0 the expected share is $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\hat{V}(\hat{P}) - \hat{V}(\hat{P}^*)}} \xrightarrow{d} N(0, 1), \quad \hat{V}(\hat{P}) = \frac{\hat{P}^*(1 - \hat{P}^*)}{n}$$

► $\hat{V}(\hat{P}^*) = [(2\hat{p}_y - 1)^2 \hat{p}_x(1 - \hat{p}_x) + (2\hat{p}_x - 1)^2 \hat{p}_y(1 - \hat{p}_y)]/n + 4\hat{p}_y \hat{p}_x(1 - \hat{p}_y)(1 - \hat{p}_x)/n^2$

□ Logit above: $\hat{p}_y = 0.581$, $\hat{p}_x = 0.983 \Rightarrow \hat{P}^* = 0.578 > \hat{P} = 0.576$; $PT = -1.28$, one-sided $p = 0.90$

□ PT conditions on the marginal frequencies automatically: no need to guess the baseline

□ Overlapping 5-day labels: PT assumes independent days; the same null as the slope t -test in $y_t = a + b x_t + u_t$ with Newey–West errors (Newey & West (1987)): $t = -1.00$

 inference_tests.py

ROC AUC and Its Sampling Uncertainty

- Receiver operating characteristic (ROC) curve: true positive rate $TP/(TP + FN)$ against false positive rate $FP/(FP + TN)$ over all thresholds
- Area under the curve (AUC) = $P(\hat{p}_{\text{pos}} > \hat{p}_{\text{neg}}) + \frac{1}{2}P(\hat{p}_{\text{pos}} = \hat{p}_{\text{neg}}) = U/(n_1 n_0)$, the Mann–Whitney statistic; 0.5 = no ranking ability
- DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)): from the placement values of each observation; assumes independent observations
- Circular block bootstrap (Künsch (1989)): resample blocks of $L = 20$ consecutive days, 1,000 replications; keeps serial dependence
- Logit, pooled out-of-fold AUC = 0.479
 - ▶ DeLong 95% CI [0.465; 0.494]: “significantly worse than random”
 - ▶ block bootstrap 95% CI [0.455; 0.504]: noise around 0.5
- An AUC below 0.5 is anti-ranking only if it survives a test that respects the dependence

Comparing Forecasts: Diebold–Mariano and Clark–West

- ▣ Loss differential $d_t = L(y_t, \hat{p}_{1,t}) - L(y_t, \hat{p}_{0,t})$, with log-loss or Brier loss $(y_t - \hat{p}_t)^2$
- ▣ Diebold–Mariano (DM): $DM = \bar{d} / \sqrt{\hat{\Omega}_{NW}/n} \xrightarrow{d} N(0, 1)$ under $H_0 : \mathbb{E}[d_t] = 0$ (Diebold & Mariano (1995)); $\hat{\Omega}_{NW}$ = heteroskedasticity and autocorrelation consistent (HAC) long-run variance
- ▣ **Nested models**: the logit with 14 features nests the constant forecast ($\beta = 0$)
 - ▶ under H_0 the large model estimates zero coefficients with noise, so $\mathbb{E}[d_t] > 0$ in finite samples
 - ▶ DM is then not $N(0, 1)$ and is undersized: it rarely finds the larger model better
- ▣ Clark–West (CW) adjustment (Clark & West (2007)): one-sided t -test (HAC) of $\mathbb{E}[f_t] = 0$, with

$$f_t = (y_t - \hat{p}_{0,t})^2 - [(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2]$$

- ▣ The last term removes the noise that a correctly nested model pays for estimating zeros
- ▣ CW derive approximate normality for nested *linear* models estimated by OLS; for penalised-logit probabilities (Brier loss) the same adjustment is an approximation, not a theorem

Conditional Predictive Ability: Giacomini–White

- **Giacomini & White (2006)**: the null concerns the forecasting **method** (model + estimation window + refitting rule), not the population model

$$H_0 : \mathbb{E}[d_{t+\tau} \mid \mathcal{F}_t] = 0 \Rightarrow \mathbb{E}[h_t d_{t+\tau}] = 0, \quad GW = n \bar{\mathbf{Z}}^\top \hat{\Omega}^{-1} \bar{\mathbf{Z}} \xrightarrow{d} \chi_q^2$$

- ▶ $\mathbf{Z}_t = h_t d_{t+\tau}$; test functions h_t (here 1 and the lagged loss differential, $q = 2$); $\tau = 5$ -day horizon
- The theory requires a bounded (rolling) estimation window, so that estimation noise never vanishes; an expanding window lies outside the theorem
- CW and GW answer different questions
 - ▶ CW: does the *population* model with features add information?
 - ▶ GW: does *my estimated* rule, as deployed, forecast better?
- Both are applied to the S&P 500 walk-forward in Section 7

From Predictions to a Trading Strategy

Definition 1 (Positions, turnover, costs, exposure)

- Position $w_t \in \{0, 1\}$: long the index (1) or in cash (0), decided with information up to t
 - Strategy return: $r_{t+1}^s = w_t r_{t+1} - c |w_t - w_{t-1}|$, with r_{t+1} the daily log return; $r - c$ is the first-order approximation of $\log(e^r - c)$
 - Turnover $|w_t - w_{t-1}|$: fraction of the portfolio traded; cost c in basis points (5 bp = 0.05% per unit traded)
 - Exposure: average share of time invested, $\bar{w} = \frac{1}{T} \sum_t w_t$
-
- Buy-and-hold: $w_t = 1$ for all t (exposure 100%, no turnover)

The Sampling Distribution of the Sharpe Ratio

- ▣ Sharpe ratio (SR) per period: $\widehat{SR} = \bar{r}/\hat{\sigma}$; in this chapter r = daily log returns (mean log growth per unit of risk), risk-free rate set to 0
- ▣ i.i.d. returns from the Normal distribution: $\text{Var}(\widehat{SR}) \approx (1 + \frac{1}{2}SR^2)/T$ (Lo (2002))
- ▣ i.i.d. non-Normal returns, delta method with skewness γ_3 and kurtosis γ_4 (Opdyke (2007)):

$$\text{Var}(\widehat{SR}) \approx \frac{1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2}{T}$$

- ▶ this is the denominator of the Probabilistic Sharpe Ratio in Section 8
- ▣ Annualising with $\sqrt{q}$ ($q = 252$ or 365) requires zero autocorrelation; otherwise $SR_q = \eta(q) SR$,
 $\eta(q) = q / \sqrt{q + 2 \sum_{k=1}^{q-1} (q-k) \rho_k}$ (Lo (2002))
- ▣ Two strategies on the same days: test $SR_a - SR_b$ with the HAC delta method or a studentised block bootstrap (Ledoit & Wolf (2008)); comparing two separate CIs ignores their correlation

Recap: Measuring and Testing Predictive Ability

- Accuracy alone is misleading with unbalanced classes; Pesaran–Timmermann conditions on the class shares
- AUC and its CI: DeLong assumes independence; a block bootstrap respects dependence
- Forecast comparison: DM for non-nested forecasts, Clark–West for nested models, Giacomini–White for estimated methods
- Sharpe ratios are estimates: standard error $\approx \sqrt{(1 + SR^2/2)/T}$; compare two with Ledoit–Wolf
- Next: how to build features and labels that do not leak information

Sampling Financial Data: Bars

Time and tick bars

- ▣ A **bar** = one row (open, high, low, close, volume) summarising a sampling interval
- ▣ Time bars: every fixed interval (1 minute, 1 day); oversample quiet periods; fat-tailed returns
- ▣ Tick bars: one bar every N transactions

Volume, dollar and information bars

- ▣ One bar every V shares or every D dollars traded: sampling by **information arrival**
- ▣ Returns closer to i.i.d. draws from the Normal distribution: [Clark \(1973\)](#); [Ané & Geman \(2000\)](#)
- ▣ Imbalance / runs bars: sample when order flow becomes one-sided

- ▣ This chapter uses daily time bars; tick data (crypto exchanges) in Chapters 10 and 16

The Stationarity vs. Memory Dilemma

- ML models need inputs whose distribution does not drift: the mapping learned in the past must remain valid
- Log prices behave as $I(1)$ (integrated of order one); returns $\Delta \log P_t$ are $I(0)$ but **erase memory** of the price level
- A series is $I(\delta)$ if $(1 - B)^\delta X_t$ is $I(0)$; it is stationary only for $\delta < \frac{1}{2}$ (Hosking (1981))
- Differencing an $I(1)$ log price by a fractional order d gives an $I(1 - d)$ series

$$0 \leq d \leq \frac{1}{2}$$

- $I(1 - d)$, $1 - d \geq \frac{1}{2}$
- non-stationary, mean-reverting for $d > 0$

$$\frac{1}{2} < d < 1$$

- stationary, long memory
- slowly decaying autocorrelations

$$d = 1 \text{ (returns)}$$

- $I(0)$
- level memory erased

- López de Prado (2018) proposes the minimum d at which an ADF test rejects; this section shows why that rule fails

Fractional Differentiation

Definition 2 (Fractional difference operator)

- With the backshift operator $BX_t = X_{t-1}$ and a real order d :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \quad \tilde{X}_t = \sum_{k=0}^{\infty} w_k X_{t-k}$$

- Weights: $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)} \sim \frac{k^{-(1+d)}}{\Gamma(-d)}$ as $k \rightarrow \infty$

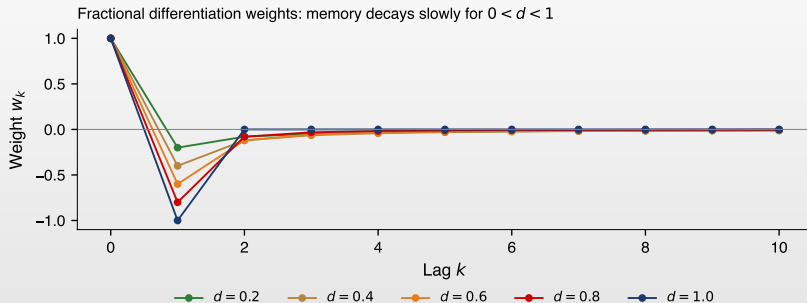
- $d = 1$: $w = (1, -1, 0, 0, \dots)$ — ordinary returns
- $0 < d < 1$: weights decay hyperbolically $\Rightarrow$ long memory is kept
- Infinite sum: $\sum_{k=0}^{\infty} w_k = (1 - 1)^d = 0$, so the level of X_t cancels only in the limit

Fixed-Width Window

Definition 3 (Fixed-width-window fractional differentiation (FFD))

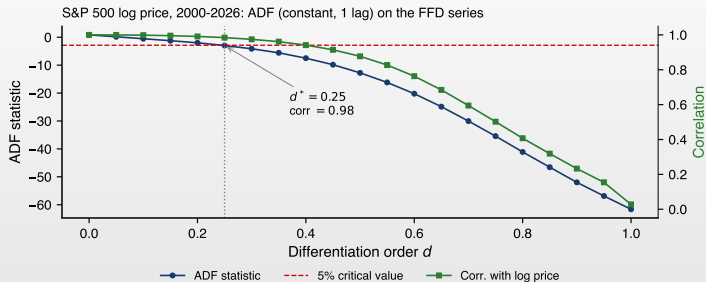
- ▣ Truncate the weights: keep $w_0, \dots, w_K$ with $|w_k| \geq \tau$ (e.g. $\tau = 10^{-4}$), drop the rest (López de Prado (2018))
- ▣ $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$ for every $t > K$
- ▣ Every $\tilde{X}_t$ uses the same number of lags; only past values enter $\Rightarrow$ no look-ahead
- ▣ The smaller d , the more lags are needed: $d = 0.25, \tau = 10^{-4} \Rightarrow K = 444$ lags
- ▣ The truncated weights no longer sum to zero: $\sum_{k=0}^K w_k = 0.178$ at $d = 0.25$ (0.040 at $d = 0.5$)

Fractional Differentiation Weights



- For $d = 1$ only the first lag matters; for small d the weights of distant lags remain non-zero
- All weights after w_0 are negative: $\tilde{X}_t$ compares today's level with a weighted history

The ADF Statistic Along d : S&P 500



- Augmented Dickey–Fuller (ADF) unit-root test (Time Series course): H_0 unit root, reject below ≈ -2.86 at 5%
- FFD log price, S&P 500 2000–2026, constant and 1 lag: the ADF rejects from $d = 0.25$, where $\text{Corr}(\tilde{X}_t, \log P_t) = 0.98$
- Same series, lag length chosen by AIC (34 lags): $\text{ADF} = -0.72$, no rejection
- A correlation of 0.98 with an $I(1)$ price is a warning, not a virtue

Truncated FFD of an $I(1)$ Price Is Still $I(1)$

- Decompose the FFD series:

$$\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k} = \left(\sum_{k=0}^K w_k \right) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$$

- If X_t is $I(1)$, each $X_t - X_{t-k}$ is $I(0)$; the first term is $0.178 X_t$ at $d = 0.25$: a **scaled random walk**
- The unit-root component has small variance next to the stationary part, so ADF with few lags confuses the two
 - ▶ Monte Carlo: pure random walk, $T = 6,274$, FFD with $d = 0.25$: ADF (1 lag) rejects in **97.8%** of 500 samples, at a nominal 5%
- Local Whittle (next slide) gives $\hat{d} = 0.75$ [0.69; 0.81] for the FFD series: the apparent memory over the band, not its order
 - ▶ above frequency $1/K$ the filter acts as $(1 - B)^{0.25}$; only $n/(2\pi K) \approx 2$ of the $m = 294$ frequencies lie below $1/K$, where the unit root returns
- Consequence: `ffd_price` in the case study is a scaled price level, not a stationary feature

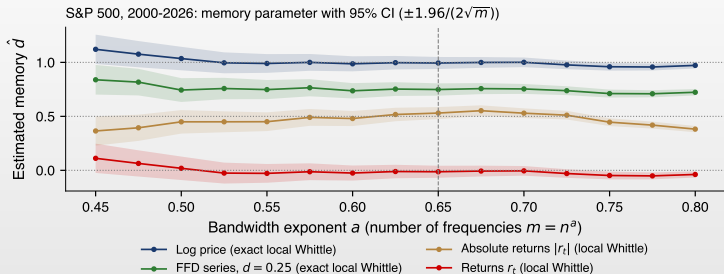
Memory Estimation Done Right: Local Whittle

- ▣ Near frequency zero an $I(\delta)$ series has spectral density $f(\lambda) \sim G \lambda^{-2\delta}$; periodogram $I(\lambda_j)$ at $\lambda_j = 2\pi j/n$
- ▣ Local Whittle (Robinson (1995)): use the first m frequencies,

$$\hat{\delta} = \arg \min_{\delta} \log \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2\delta} I(\lambda_j) \right) - \frac{2\delta}{m} \sum_{j=1}^m \log \lambda_j, \quad \sqrt{m}(\hat{\delta} - \delta) \xrightarrow{d} N(0, \frac{1}{4})$$

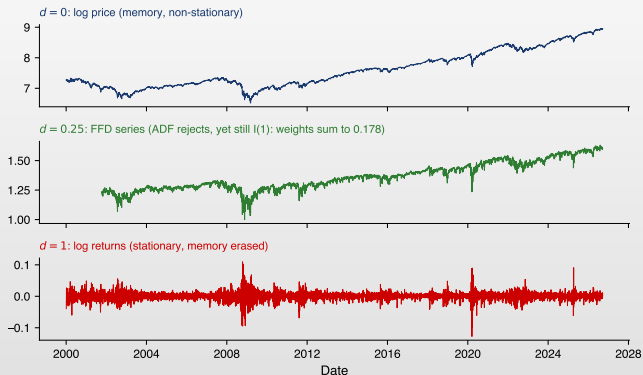
- ▣ Exact local Whittle (ELW, Shimotsu & Phillips (2005); unknown mean: Shimotsu (2010)): periodogram of $(1 - B)^{\delta} X_t$, valid also for non-stationary $\delta \geq \frac{1}{2}$
- ▣ S&P 500, 2000–2026, $m = n^{0.65}$, 95% CI $\pm 1.96/(2\sqrt{m})$
 - ▶ log price (ELW): 0.99 [0.94; 1.05]; returns: -0.01 [-0.07; 0.04]
 - ▶ absolute returns $|r_t|$: 0.53 [0.47; 0.59]; FFD series, $d = 0.25$ (ELW): 0.75 [0.69; 0.81]
- ▣ The long memory is in **volatility**, not in the price level: fractional features belong on $|r_t|$ (Franke, Härdle & Hafner (2019))

Memory Estimates Across Bandwidths



- Bandwidth $m = n^a$: small a = less bias from short-run dynamics, wider CI; report the whole curve
- Log price stays at 1; the FFD series stays near $1 - d$ because even small bandwidths use few frequencies below $1/K$ (its order is still 1): neither is stationary
- Returns: no memory; absolute returns: $\hat{d} \approx 0.36\text{--}0.55$, the long memory of volatility (Chapters 5 and 9)

Prices, FFD Series and Returns



- ▣ **Top:** log price — $I(1)$
- ▣ **Middle:** FFD with $d = 0.25$ — the upward trend of the price is still visible: $0.178 \log P_t$ plus a stationary part
- ▣ **Bottom:** log returns — only volatility clusters are visible
- ▣ The FFD series enters the case study as feature `ffd_price`

Choosing d on the Full Sample Is Leakage

- The “minimum d ” was chosen on 2000–2026 and then used inside purged CV: the test folds helped choose a feature (Section 6)
 - ▶ the same ADF rule on data up to 2009 gives $d = 0.15$; on the full sample $d = 0.25$
- Case study without `ffd_price` (purged 5-fold, mean over folds)
 - ▶ accuracy: logit 57.5%, RF 56.4%, GBM 55.6%, MLP 55.5%, all below 58.1%
 - ▶ AUC: 0.483, 0.522, 0.514, 0.514
- The negative conclusion does not depend on this feature
- Correct protocol
 - ▶ estimate d (local Whittle, with its CI) inside each training fold only
 - ▶ prefer features with an economic meaning: distances to moving averages, fractional filters of $|r_t|$

 `inference_tests.py`

Recap: Financial Data and Features

- ▣ Sampling by activity (volume, dollar bars) gives better-behaved returns than clock time
- ▣ An ADF rejection is not stationarity: truncated FFD of an $I(1)$ price stays $I(1)$ (weights sum to 0.178 at $d = 0.25$)
- ▣ Estimate memory directly: local Whittle / ELW with a CI; S&P 500 log price $\hat{d} = 0.99$, absolute returns 0.53
- ▣ Every transformation, including the choice of d , uses only past data and training folds
- ▣ Next: how to turn prices into labels

The Fixed-Horizon Label and Its Problems

Definition 4 (Fixed-horizon label)

- h -period log return $r_{t,t+h} = \log P_{t+h} - \log P_t$ and threshold $\tau \geq 0$: $y_t = +1$ if $r_{t,t+h} > \tau$, -1 if $r_{t,t+h} < -\tau$, 0 otherwise
- **Label shares**: fraction of observations in each class, e.g. $\frac{1}{n} \sum_t 1\{y_t = +1\}$

Problems

- A constant τ ignores time-varying volatility: in calm periods almost all labels are 0, in crises almost none
- It ignores the **path**: a real position would have been stopped out long before $t + h$
- Unbalanced classes: S&P 500, $h = 10$ days, $\tau = 0$: 60.5% up vs. 39.5% down

The Triple-Barrier Method

Definition 5 (Triple-barrier label, López de Prado (2018))

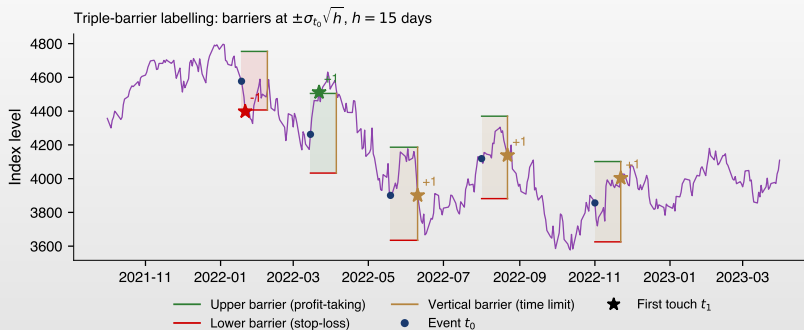
- At event time t_0 , with daily volatility σ_{t_0} and horizon h :
 - ▶ upper barrier (profit-taking): $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$
 - ▶ lower barrier (stop-loss): $\log P_t - \log P_{t_0} \leq -\pi_l \sigma_{t_0} \sqrt{h}$
 - ▶ vertical barrier (time limit): $t = t_0 + h$
- Label = **first barrier touched** at time t_1 : +1 upper, -1 lower; at the vertical barrier either $\text{sign}(r_{t_0, t_1})$ or 0
- π_u, π_l : barrier widths in units of $\sigma_{t_0} \sqrt{h}$

Volatility Estimate and Properties of the Barriers

Definition 6 (EWMA volatility)

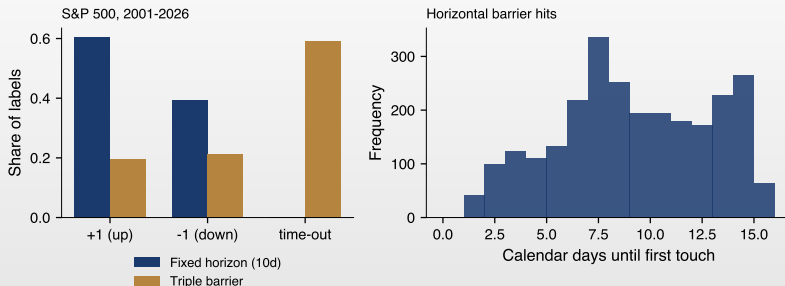
- Square root of the exponentially weighted moving average (EWMA) of squared deviations of daily log returns, with span 50 (decay $\lambda = 1 - 2/51$): recent days weigh more
- Barriers scale with volatility $\Rightarrow$ labels are comparable across calm and turbulent regimes
- Mimics how a trader manages a position: take-profit, stop-loss, maximum holding period
- Each label has a **span** $[t_0, t_1]$ — essential for cross-validation (Section 6)

Triple Barrier: S&P 500 in 2022



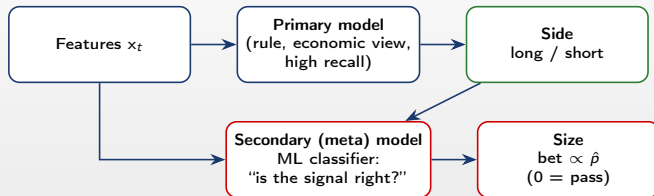
- Five events during the 2022 bear market, $\pi_u = \pi_l = 1$, $h = 15$ days
- January: stop-loss hit (-1); March: profit-taking ($+1$); the others end at the vertical barrier

Label Distribution: Fixed Horizon vs. Triple Barrier



- S&P 500, 2001–2026, $h = 10$: fixed horizon gives 60.5% up / 39.5% down
- Triple barrier ($\pm 1\sigma\sqrt{h}$): 19.6% up, 21.3% down, 59.0% time-out — balanced directional labels
- Right panel: horizontal barriers are typically hit after 1–2 weeks

Meta-Labeling



- Meta-label: $y_t^{meta} = 1$ if the primary signal was profitable (barrier touched in its direction), 0 otherwise
- The secondary model outputs $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$

Why Meta-Labelling Works

- ▣ ML learns **when to trust** the primary model, not the direction itself
- ▣ The primary model is tuned for high recall (catch most opportunities)
- ▣ The meta model filters out false positives $\Rightarrow$ higher precision, and a better F1 score when recall is not sacrificed too much
- ▣ Keeps an interpretable, economically motivated core; limits overfitting of the direction
- ▣ The seminar applies it to a moving-average crossover on Bitcoin

Overlapping Labels and Sample Uniqueness

- Label i spans $[t_{0,i}, t_{1,i}]$; consecutive labels share returns $\Rightarrow$ samples are **not i.i.d.**
- Concurrent labels at t : $c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$; average uniqueness $\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$
- $\bar{u}_i = 1$: no shared return; $\bar{u}_i = 1/5$: the harmonic mean of c_t over the span of label i is five

Consequences

- Effective sample size $\approx \sum_i \bar{u}_i \ll n$ (an overlap heuristic, not a variance-based measure)
- Bagging draws near-duplicates; CV scores too optimistic

Remedies

- Weight samples by $\bar{u}_i$ (and by $|r|$)
- Sequential bootstrap; in RF `max_samples` $\approx \bar{u}$

Recap: Labelling

- ▣ Fixed-horizon labels ignore volatility and the price path, and are unbalanced (60.5% up)
- ▣ Triple-barrier labels scale with volatility and mimic real trade management
- ▣ Meta-labelling lets ML decide *whether* to act on an economic signal
- ▣ Every label has a span; overlapping spans shrink the effective sample
- ▣ Next: validation that respects these spans

Information Leakage: Why Standard K-Fold Fails

Definition 7 (Leakage)

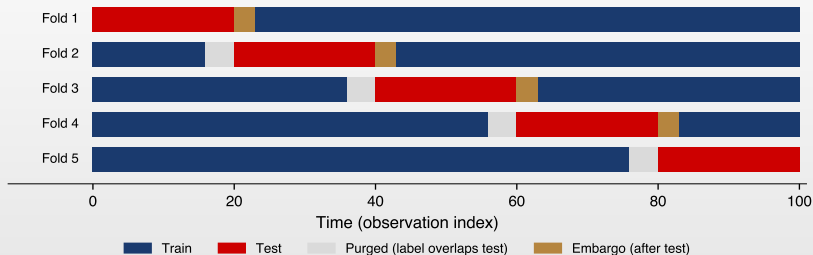
- The training set contains information that is also present in the test set, so the test score no longer measures performance on genuinely new data
- Serially correlated features ($x_t \approx x_{t+1}$) and overlapping labels ($y_t \approx y_{t+1}$): a shuffled K -fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour $\Rightarrow$ excellent “out-of-sample” score; tuning on a leaky CV selects the **most overfitted** model
- Other sources: transformations fitted on the full sample (scaling, principal component analysis, the choice of d for FFD), survivorship bias, data revisions, wrong timestamps

Purging and Embargo

Definition 8 (Purged K-fold with embargo, López de Prado (2018))

- ▣ Folds are **contiguous** blocks in time (no shuffling)
 - ▣ **Purging**: remove from training every observation whose label span $[t_{0,i}, t_{1,i}]$ overlaps the test period
 - ▣ **Embargo**: also remove a small fraction (e.g. 1%) of observations right after the test block, because features look back in time
- ▣ Purging handles label overlap; the embargo handles feature memory

Purged K-Fold: The Scheme



- Red: test block; grey: purged observations just before it; amber: embargo just after it
- The training set shrinks a little — the price of an honest estimate

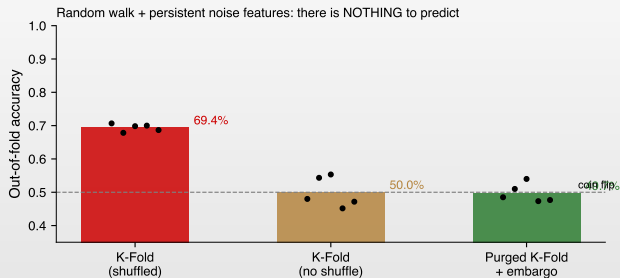
Worked Example: Purging with 10 Observations

- Observations $t = 0, \dots, 9$; label of t ends at $t + 2$ (capped at 9)
- 5 contiguous folds of 2; embargo = 1 observation

Test	Removed	Train
0, 1	2, 3	4–9
2, 3	0, 1, 4, 5	6–9
4, 5	2, 3, 6, 7	0, 1, 8, 9
6, 7	4, 5, 8, 9	0–3
8, 9	6, 7	0–5

- Test $\{4, 5\}$: labels of 2 and 3 end at 4 and 5 $\Rightarrow$ purged; 6 is the embargo; 7 starts when the last test label ends $\Rightarrow$ purged
- With overlapping labels, up to 4 of 10 observations leave the training set

Leakage Experiment: Nothing to Predict



- Price = pure random walk; label = sign of the 20-day forward return
- 5 features = rolling means of independent noise
- True accuracy is 50% by construction
- Random forest classifier, 5 folds

 MFM_ch13_purged_cv

Leakage Experiment: Results

- ▣ **Shuffled K-fold: 69.4%** accuracy — pure leakage
- ▣ Unshuffled (contiguous) K-fold: 50.0%
- ▣ Purged K-fold + embargo: 49.7%

Lesson

- ▣ A 69% hit rate on data with zero signal
- ▣ Shuffling fails here because labels overlap *and* features persist (next slide: when it is harmless)
- ▣ A result that looks too good is usually leakage

When Is K-Fold Valid for Time Series?

- Bergmeir, Hyndman & Koo (2018): for a **purely autoregressive** model $y_t = g(y_{t-1}, \dots, y_{t-p}) + \varepsilon_t$ with serially uncorrelated errors, standard K -fold CV is valid
 - ▶ the errors in the test fold are uncorrelated with the training errors, so the CV error is unbiased for the OOS error
- It fails when the **errors** are serially correlated
 - ▶ overlapping labels: h -day forward returns share $h - 1$ daily returns
 - ▶ persistent features that are not in the model's lag set (rolling means, levels such as `ffd_price`)
 - ▶ under-specified dynamics (too few lags)
- The leakage experiment isolates exactly this: 20-day labels and 20-day rolling features
- Check before choosing: the autocorrelation of the CV residuals; if it is significant, purge and embargo

Walk-Forward and Combinatorial Purged CV

Walk-forward (WF)

- Train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward
- Mimics real-time deployment; no future data in training
- But a **single path**: high variance, easy to overfit by repeated trials

Combinatorial purged CV (CPCV)

- N groups; test on all $\binom{N}{k}$ combinations of k groups, with purging and embargo
- $\varphi = \frac{k}{N} \binom{N}{k}$ backtest paths; e.g. $N = 6$, $k = 2$: 15 splits, 5 paths
- A **distribution** of Sharpe ratios instead of one number

- Rule of thumb: purged K-fold / CPCV for model selection; WF for the final, realistic evaluation

Recap: Cross-Validation in Finance

- ▣ Leakage comes from overlapping labels and serially correlated features
- ▣ Shuffled K-fold reports 69.4% on pure noise; purged K-fold reports 49.7%
- ▣ K-fold is valid for autoregressions with uncorrelated errors; overlapping labels break it
- ▣ Purging removes overlapping labels; the embargo removes the feature memory after the test block
- ▣ Walk-forward = realistic single path; CPCV = many paths, a distribution of results
- ▣ Next: all of this applied to the S&P 500

Case Study: Data, Target and Validation

- ▣ S&P 500 index and VIX, daily, 2001–2026: 6,269 observations
- ▣ Target: $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$ — the 5-day direction
- ▣ Label span $[t, t + 5]$ used for purging
- ▣ Purged 5-fold CV with 1% embargo
- ▣ Metrics: accuracy and ROC AUC, compared with the “always up” baseline

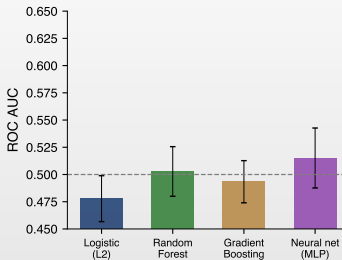
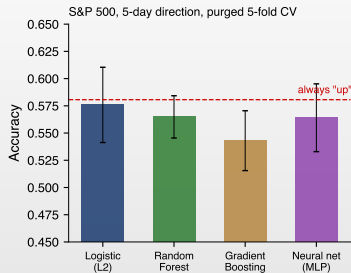
Case Study: 14 Features

- ▣ **Momentum:** $\log P_t - \log P_{t-h}$, $h \in \{1, 5, 20, 60, 120\}$ days
- ▣ **Realised volatility:** annualised standard deviation of daily log returns over 20 and 60 days, and their ratio
- ▣ **Distance to moving average:** $\log P_t$ minus the mean of $\log P$ over the last 50 or 200 days
- ▣ **RSI(14):** $100 - 100/(1 + RS)$, RS = ratio of average gains to average losses over 14 days (smoothed)
- ▣ **VIX:** log level and 5-day change of the implied-volatility index
- ▣ **FFD** log price with $d = 0.25$: a scaled price level (Section 4)
- ▣ All features use information up to day t ; only the order d of the FFD feature was chosen on the full sample (Section 4)

Case Study: Four Models

- ▣ **Logit:** L2-penalised (Ridge) logistic regression on standardised features
- ▣ **RF:** 400 trees, minimum leaf size 100
- ▣ **GBM:** histogram-based, tree depth 3, learning rate 0.03, 200 trees
- ▣ **MLP:** two hidden layers (32 and 16 units), early stopping
- ▣ Settings deliberately conservative: large leaves and shallow trees limit variance

Model Comparison



	Logit	RF	GBM	MLP
Accuracy	57.6%	56.5%	54.3%	56.4%
AUC	0.478	0.503	0.493	0.515

□ Baseline “always up”: 58.1%

□ No model beats it

Case Study: Are the Differences Significant?

	Logit	RF	GBM	MLP
Accuracy $\hat{P}$ (pooled)	0.576	0.565	0.543	0.564
Expected under independence $\hat{P}^*$	0.578	0.570	0.557	0.565
Pesaran–Timmermann PT	-1.28	-1.61	-3.07	-0.36
HAC slope t (lag 4)	-1.00	-1.16	-2.31	-0.26
Pooled AUC	0.479	0.495	0.490	0.512
Block-bootstrap 95% CI	[0.455; 0.504]	[0.472; 0.520]	[0.468; 0.514]	[0.489; 0.537]

- No model shows significant positive directional skill; every AUC interval contains 0.5
- GBM is significantly *worse* than independence ($PT = -3.07$, HAC $t = -2.31$): it predicts “down” at the wrong times
- Pooled AUCs differ slightly from the fold means of the previous slide: one ranking over all days vs. the average of five

Interpreting the Results

Honest conclusion

- ▣ **No model beats the naive “always up” baseline**; all AUC intervals contain 0.5
- ▣ Accuracy is misleading with unbalanced classes: 57.6% looks good but is *below* 58.1%
- ▣ Consistent with (weak-form) efficiency of a highly liquid index (Chapter 2)

Where to look instead

- ▣ Cross-sections of many assets (Gu, Kelly & Xiu (2020)); volatility and risk targets; crypto delisting risk (Będowska-Sójka, Wójcik & Pele (2026))
- ▣ Alternative data: text (Chapter 15); less efficient markets: crypto (Chapter 16)
- ▣ Meta-labelling of economically motivated signals

Feature Importance: MDI and MDA

Mean decrease impurity (MDI)

- Average reduction of Gini impurity from all splits on a feature, over the trees of a forest
- In-sample, trees only
- Biased towards continuous, high-cardinality features

Mean decrease accuracy (MDA)

- Permutation importance: drop in an OOS score (here AUC) when the values of a feature are randomly shuffled
- Any model; out of sample
- Can be negative: the feature hurts

- MDI answers “what did the model use?”; MDA answers “what helps on new data?”

Feature Importance: SHAP

Definition 9 (Shapley value of feature j (SHAP, Lundberg & Lee (2017)))

- Average marginal contribution of j over all coalitions $S \subseteq \{1, \dots, p\} \setminus \{j\}$:

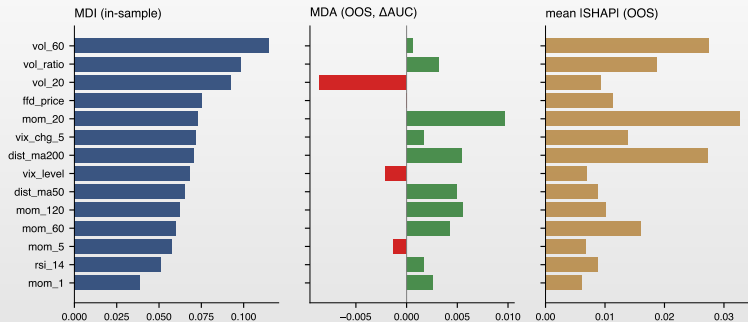
$$\phi_j = \sum_S \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)], \quad \hat{f}(x) = \phi_0 + \sum_j \phi_j$$

- Interventional SHAP: $v(S) = \mathbb{E}[\hat{f}(x_S, X_{\bar{S}})]$, averaging $\hat{f}$ with the features outside S drawn from a background sample (as in the Quantlet); $\phi_0 = v(\emptyset) =$ average prediction

- Conditional SHAP uses $v(S) = \mathbb{E}[\hat{f}(X) \mid X_S = x_S]$; the two differ when features are dependent
- Additive attribution of each prediction; mean $|\phi_j|$ over observations = global importance
- **Substitution effects**: correlated features share (and dilute) importance — cluster them first



Feature Importance in the Case Study



- Three panels: MDI (in-sample), MDA (ΔAUC out of sample), mean |SHAP| (out of sample)
- MDI, MDA: random forest; SHAP: gradient boosting, background of 200 training days; fit on the first 70%, evaluated on the last 30% after a purge

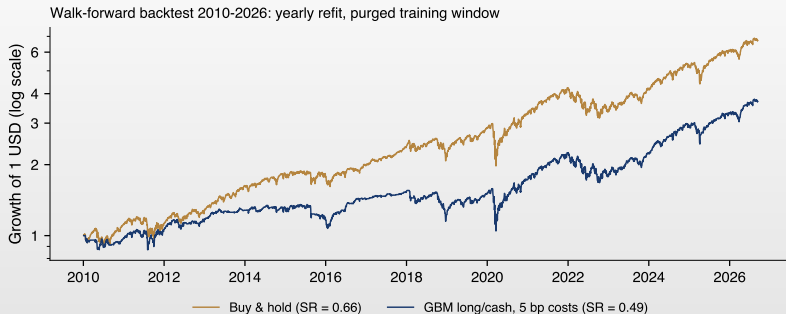
Feature Importance: Interpretation

- ▣ **MDI** ranks volatility features first; **MDA** and **SHAP** point to `mom_20` and `dist_ma200`
- ▣ `vol_20`: high MDI but **negative** MDA — a sign of overfitting, not a proof: sampling noise and correlated features also move MDA
- ▣ Importances are tiny ($\Delta\text{AUC} < 0.01$), consistent with the model comparison
- ▣ Permuting one of several correlated features creates points off the data manifold: MDA and SHAP then **extrapolate** (Hooker, Mentch & Zhou (2021)); permute within clusters or refit without the feature
- ▣ Importances are estimates and need their own uncertainty analysis; honest trees give asymptotically Normal forest *predictions* and CIs (Wager & Athey (2018)), not CIs for importances
- ▣ **Backtesting is not a research tool**, importance analysis is: López de Prado (2018)

Walk-Forward Backtest: Design

- Model: GBM refitted every January on all past data, with purging of labels that overlap the test year
- Rule: long the S&P 500 if $\hat{p} > 0.5$, otherwise cash ($w_t \in \{0, 1\}$)
- Costs: 5 bp per unit of turnover
- Out-of-sample period: 2010–2026, one refit per year
- Benchmark: buy-and-hold ($w_t = 1$)

Walk-Forward Backtest: Results



- 2010–2026: strategy Sharpe **0.49** vs. buy-and-hold **0.66** (annualised); average exposure 77%
- The model mostly learned “be long”, and its exits came at the wrong times
- $d = 0.25$ of `ffd_price` was chosen on 2000–2026; with $d = 0.15$, chosen by the same rule on data up to 2009, SR = 0.43; without `ffd_price` 0.51

Walk-Forward: Is the Gap Significant?

- ▣ Walk-forward 2010–2026, $n = 4,198$ days, 60.2% “up”; benchmark: constant forecast = training share of “up”
- ▣ Logit (nests the constant): log-loss 0.6822 vs. 0.6757
 - ▶ DM $t = 2.39$ ($p = 0.017$): the estimated logit is significantly *worse*
 - ▶ Clark–West $t = -0.21$ (one-sided $p = 0.58$): no evidence that the features carry information
 - ▶ Giacomini–White $= 5.77$ (χ^2_2 , $p = 0.056$): no significant difference at 5%; approximate, since the window expands
 - ▶ rolling 8-year window, as the GW theory requires: GW $= 26.5$ ($p < 0.001$): the logit forecasts worse than the constant
- ▣ GBM: log-loss 0.6971; DM $t = 5.09$, GW $= 25.8$ ($p < 0.001$): worse than a constant; Pesaran–Timmermann -0.74
- ▣ Sharpe 0.49 vs. 0.66 (Lo SE ≈ 0.25 each): difference -0.17 , Ledoit–Wolf HAC SE 0.095, $p = 0.073$; block bootstrap CI $[-0.36; 0.02]$
- ▣ The CI contains 0: no evidence of skill, and no significant evidence that the strategy is worse than buy-and-hold

Recap: Case Study

- ▣ Four models, 14 features, 25 years, purged validation; d of the FFD feature was chosen on the full sample, and dropping it changes nothing
- ▣ No model beats “always up” (58.1%); Pesaran–Timmermann finds no significant directional skill; every AUC CI contains 0.5
- ▣ Feature importances are tiny and differ between in-sample and out-of-sample measures
- ▣ Walk-forward: the models lose to a constant in log-loss (DM, GW); Sharpe 0.49 vs. 0.66, $p = 0.073$
- ▣ An honest negative result is a result

Multiple Testing and Selection Bias

- ▣ A **backtest**: simulation of a strategy on historical data
- ▣ A researcher tries N strategies or configurations and reports the best one
- ▣ Even if all have zero true Sharpe ratio, $\max_n \widehat{SR}_n$ grows with N
- ▣ **Selection bias**: the reported maximum overstates the true performance
- ▣ [Harvey, Liu & Zhu \(2016\)](#): given hundreds of published factors, a new factor needs a t -statistic above 3, not 2

The False Strategy Theorem

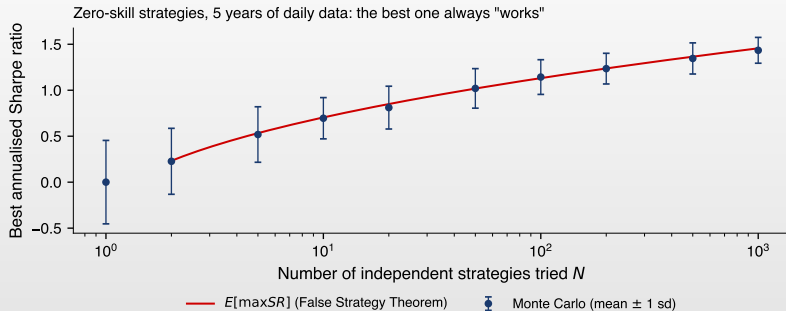
Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- Let $\widehat{SR}_n$, $n = 1, \dots, N$, be independent estimates, approximately $N(0, \text{Var}[\widehat{SR}])$ (no skill); then the **expected maximum Sharpe ratio** is

$$\mathbb{E}\left[\max_n \widehat{SR}_n\right] \approx \sqrt{\text{Var}[\widehat{SR}]} \left[(1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{N_e}\right) \right]$$

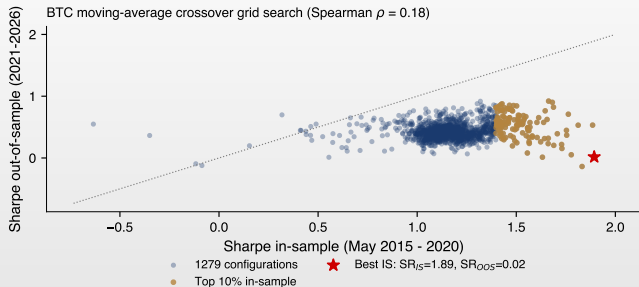
- $\gamma \approx 0.5772$: Euler–Mascheroni constant; Φ : standard Normal cumulative distribution function
- For i.i.d. returns, $\text{Var}[\widehat{SR}] \approx 1/T$ per period

The Best of N Zero-Skill Strategies



- Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised $\text{Var}[\widehat{SR}] \approx 1/5$)
- Trying $N = 1000$ variants yields a best Sharpe ratio of ≈ 1.45 — **with no skill at all**
- The Monte Carlo simulation agrees with the theorem

In-Sample vs. Out-of-Sample: Bitcoin Grid Search



- Moving-average (MA) crossover on Bitcoin: long when the fast MA of the log price exceeds the slow MA by more than a band
- 1,279 configurations (fast window, slow window, band); in-sample 15 May 2015 – 2020, out-of-sample 2021–2026
- Every rule starts on the first day on which the slowest (240-day) average exists

Bitcoin Grid Search: Interpretation

- ▣ Best IS configuration (14/20 days): $SR_{IS} = 1.89 \rightarrow SR_{OOS} = 0.02$ (annualised with $\sqrt{365}$)
- ▣ Spearman rank correlation between IS and OOS Sharpe ratios: $\rho = 0.18$
- ▣ Median OOS SR: 0.55 for the top-10% IS configurations vs. 0.42 for all
- ▣ The *family* of trend rules carries some information, but the *selected* optimum is noise

Probabilistic Sharpe Ratio

Definition 10 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

- Observed per-period $\widehat{SR}$ from T returns, skewness $\hat{\gamma}_3$, kurtosis $\hat{\gamma}_4$ (Normal distribution: $\gamma_4 = 3$):

$$\widehat{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- Skewness $\gamma_3 = \mathbb{E}[(r - \mu)^3]/\sigma^3$; kurtosis $\gamma_4 = \mathbb{E}[(r - \mu)^4]/\sigma^4$ (Chapter 1)
- PSR = probability that the true SR exceeds a benchmark SR^* , given sample length and non-Normality
- The denominator is $\sqrt{T \text{Var}(\widehat{SR})}$ from the delta method of Section 3

Deflated Sharpe Ratio

Definition 11 (Deflated Sharpe Ratio (DSR), Bailey & López de Prado (2014))

□ $DSR = \widehat{PSR}(SR_0)$, with the benchmark set to the expected maximum under the null of no skill:

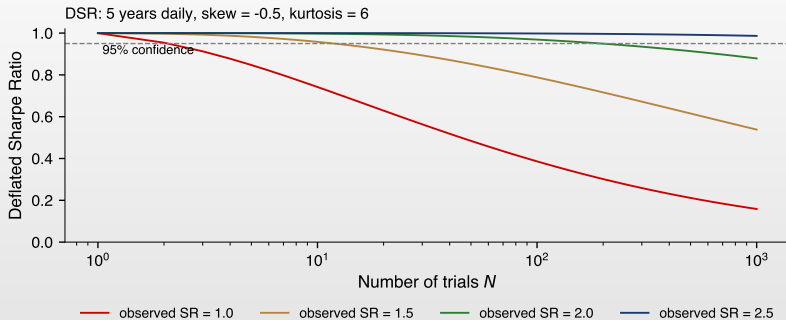
$$SR_0 = \sqrt{\text{Var}[\widehat{SR}_n]} \left[(1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{N_e}\right) \right]$$

- DSR corrects for **selection bias** (N trials), **non-Normality** and **sample length**
- Reject the strategy unless $DSR > 0.95$; report N honestly

Worked Example: DSR of the Best Bitcoin Rule

- ▣ Best IS rule: $T = 2,058$ daily returns, $\widehat{SR} = 0.0991$ per day (1.89 annualised), $\hat{\gamma}_3 = 0.12$, $\hat{\gamma}_4 = 12.33$
- ▣ Denominator: $\sqrt{1 - 0.12 \cdot 0.0991 + \frac{12.33-1}{4} \cdot 0.0991^2} = 1.008$
- ▣ $N = 1,279$ trials. Two choices of $\text{Var}[\widehat{SR}_n]$:
 - ▶ dispersion of the 1,279 trial SRs (std. 0.0106): $SR_0 = 0.035/\text{day}$, $z = 2.88$, $DSR = 0.998$
 - ▶ independent-trials value $1/T$ (std. 0.0220): $SR_0 = 0.073/\text{day}$, $z = 1.16$, $DSR = 0.877 < 0.95$
- ▣ The trials are highly correlated (neighbouring windows), so their dispersion understates the true search: the verdict depends on this choice
- ▣ The out-of-sample Sharpe of 0.02 sides with the conservative value

Deflated Sharpe Ratio in Practice



- 5 years of daily returns, skewness -0.5 , kurtosis 6
- An annualised SR of 1.0 is significant only if (almost) no other strategy was tried: it fails after 2–3 trials
- $SR = 1.5$ survives ≈ 10 trials, $SR = 2.0 \approx 200$; only $SR = 2.5$ survives 1000

Data-Snooping Tests: Reality Check, SPA, Romano–Wolf

- K rules, performance differentials $d_{k,t} = r_{k,t} - r_{0,t}$ against a benchmark r_0 ; $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$
- Reality Check (RC, [White \(2000\)](#)): statistic $\max_k \sqrt{n} \bar{d}_k$; null distribution from the **stationary bootstrap** ([Politis & Romano \(1994\)](#)) of the whole $n \times K$ matrix
 - ▶ resampling whole days keeps the correlation between rules: no need to count “effective” trials
- Superior Predictive Ability (SPA, [Hansen \(2005\)](#)): studentised $\max_k \sqrt{n} \bar{d}_k / \hat{\omega}_k$; clearly bad rules are not recentred, so adding junk rules does not destroy power
- Romano–Wolf StepM ([Romano & Wolf \(2005\)](#)): step-down procedure that names *which* rules beat the benchmark, with family-wise error rate (FWER) control
- DSR asks “is the best SR above the luck of N trials?” and needs N and $\text{Var}[\widehat{SR}]$; these tests take both from the data

SPA vs. DSR on the Bitcoin Grid

- ▣ 1,279 moving-average rules, in-sample period, $n = 2,058$ days; stationary bootstrap, mean block 20 days, 2,000 replications
- ▣ Benchmark **cash** ($r_0 = 0$): RC $p = 0.004$, SPA $p = 0.002$; StepM keeps 314 rules
 - ▶ some trend rule earned a positive mean: Bitcoin's own trend, shared by the whole grid
- ▣ Benchmark **buy-and-hold**: largest studentised statistic 0.88 (the 14/20 rule); RC $p = 0.74$, SPA $p = 0.70$; StepM keeps 0 rules
 - ▶ no rule significantly beats simply holding Bitcoin, even in sample
- ▣ DSR of the same best rule: 0.998 or 0.877 depending on $\text{Var}[\widehat{SR}]$ (worked example); SPA needs no such choice
- ▣ Out of sample the winner earned SR 0.02: consistent with the SPA verdict against buy-and-hold

 [inference_tests.py](#)

Probability of Backtest Overfitting (PBO)

Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix M of returns: T periods $\times$ N strategy configurations
2. Split rows into S blocks; for each of the $\binom{S}{S/2}$ combinations, form IS (half the blocks) and OOS (the other half)
3. Select the best configuration n^* in-sample; compute its relative rank $\bar{\omega} \in (0, 1)$ out-of-sample
4. Logit $\lambda = \log \frac{\bar{\omega}}{1-\bar{\omega}}$
5. $PBO = \Pr[\lambda < 0]$: probability that the IS winner ranks below the OOS median

Reading the PBO

- ▣ $PBO \approx 0$: the in-sample winner reliably stays above the median out of sample
- ▣ $PBO \approx 0.5$ or higher: the selection process has no skill — choosing the IS winner is like choosing at random
- ▣ PBO complements DSR: it evaluates the **research procedure**, not a single strategy
- ▣ Both require logging **every** trial, including the failed ones

Recap: Backtest Overfitting

- ▣ The best of many backtests is biased upwards: 1000 zero-skill trials give a Sharpe of ≈ 1.45
- ▣ Bitcoin grid: IS winner 1.89 $\rightarrow$ OOS 0.02
- ▣ PSR corrects for sample length and non-Normality; DSR also for the number of trials
- ▣ RC / SPA / StepM bootstrap the whole grid: against buy-and-hold no Bitcoin rule survives ($p = 0.70$)
- ▣ PBO measures the reliability of the whole selection procedure
- ▣ Record N ; deflate before you claim

LASSO Selection Is Not Inference

- The LASSO path (Section 2) answers “which feature enters first”, not “which feature matters”
- OLS t -statistics after LASSO selection ignore the selection step: they are over-optimistic and not uniformly valid
- A control that is correlated with the regressor of interest d_t but only weakly with y_t gets dropped
 - ▶ its effect moves into the coefficient of d_t : omitted-variable bias
- Worked example (below): LASSO of the 5-day S&P 500 return on log VIX and 90 controls keeps **no** regressor
 - ▶ “the VIX does not matter”, reported with no standard error at all
- Remedies: sample splitting, post-double selection, double/debiased machine learning (DML)

Post-Double Selection

- Partially linear model: $y_t = \theta d_t + g(x_t) + \varepsilon_t$, $d_t = m(x_t) + v_t$; $\mathbb{E}[\varepsilon_t | d_t, x_t] = 0$, $\mathbb{E}[v_t | x_t] = 0$, $\text{Var}(v_t) > 0$; g , m approximately sparse
- Belloni, Chernozhukov & Hansen (2014):
 1. LASSO of y_t on $x_t \Rightarrow$ selected set $\hat{S}_y$
 2. LASSO of d_t on $x_t \Rightarrow$ selected set $\hat{S}_d$
 3. OLS of y_t on d_t and $x_{\hat{S}_y \cup \hat{S}_d}$; its heteroskedasticity-robust standard error is valid under independent sampling, regularity conditions and sparsity $s^2 \log^2(p \vee n)/n \rightarrow 0$; with time series, HAC errors are an approximation
- LASSO objective $\frac{1}{n} \sum_t (y_t - x_t^\top \beta)^2 + \frac{\lambda}{n} \sum_j \hat{\psi}_j |\beta_j|$, plug-in $\lambda = 2c\sqrt{n} \Phi^{-1}(1 - \gamma/(2p))$, $c = 1.1$, $\gamma = 0.1/\log n$, loadings $\hat{\psi}_j^2 = \frac{1}{n} \sum_t x_{tj}^2 \hat{e}_t^2$
- Why it works: a control is missed only if it matters for *neither* equation, so the omitted bias is second order

Double/Debiased ML: Orthogonal Scores and Cross-Fitting

- ▣ Chernozhukov et al. (2018): residualise both variables with *any* learner (LASSO, RF, boosting)

$$\tilde{y}_t = y_t - \hat{\ell}(x_t), \quad \tilde{v}_t = d_t - \hat{m}(x_t), \quad \hat{\theta} = \frac{\sum_t \tilde{v}_t \tilde{y}_t}{\sum_t \tilde{v}_t^2}$$

- ▣ Neyman-orthogonal score $\psi = (\tilde{y} - \theta \tilde{v}) \tilde{v}$: first-order errors in $\hat{\ell}$, $\hat{m}$ do not bias $\hat{\theta}$; with $\ell(x) = \mathbb{E}[y \mid x]$, it suffices (with consistency and regularity conditions) that $\|\hat{m} - m\| (\|\hat{\ell} - \ell\| + \|\hat{m} - m\|) = o_P(n^{-1/2})$, e.g. both $o_P(n^{-1/4})$
- ▣ Cross-fitting: fit $\hat{\ell}$, $\hat{m}$ on $K - 1$ folds, residualise the held-out fold, repeat; removes the own-observation overfitting bias
- ▣ The published theory assumes independent sampling
 - ▶ with overlapping 5-day labels we use contiguous folds purged by 5 days around the test fold (Section 6) and HAC standard errors
 - ▶ a data limitation stated as such, not a new estimator

Worked Example: Does the VIX Predict S&P 500 Returns?

- y_t : 5-day forward log return (bp); d_t : log VIX, standardised (1 SD = 0.356)
- Controls: 12 technical features, their squares and pairwise products, $p = 90$; $n = 6,269$
- HAC standard errors (4 lags); DML with 5 purged contiguous folds

Method	$\hat{\theta}$ (bp)	SE
OLS, VIX only	7.5	9.0
OLS, all 90 controls	-0.7	16.1
LASSO on (VIX, controls)	dropped	—
Post-double selection	9.4	16.1
DML, LASSO learners	4.8	19.7
DML, random forests	-11.7	16.7

- Double selection keeps $|\hat{S}_y| = 0$ and $|\hat{S}_d| = 15$ controls: the ones that predict the VIX, which a single LASSO drops
- Post-double selection 95% CI $[-22.1; 40.8]$ bp per week per SD of log VIX
- Reading: the single LASSO claims “no effect” with false certainty; the honest intervals contain zero and economically large positive and negative predictive effects of log VIX

Machine Learning for Inference in Asset Pricing

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add pricing information given hundreds of existing ones?
 - ▶ double selection of the controlling factors from the zoo, then a valid t -test of the new factor's risk price
 - ▶ the inferential answer to [Harvey, Liu & Zhu \(2016\)](#), who raise the hurdle to $t > 3$
- Heterogeneous effects: honest causal forests give CIs for effects that vary with covariates ([Wager & Athey \(2018\)](#))
- Three rules
 - ▶ prediction and inference are different targets: a good predictor can give a biased $\hat{\theta}$
 - ▶ orthogonalise before you test, and cross-fit with folds that respect time
 - ▶ report CIs; “selected” or “important” is not “significant”

Recap: Machine Learning for Inference

- ▣ Regularised selection answers a prediction question; its coefficients have no valid standard errors
- ▣ Post-double selection and DML give valid inference on one parameter with many controls, under sparsity or rate conditions and independent sampling
- ▣ Cross-fitting must respect time: contiguous, purged folds and HAC errors
- ▣ VIX and 5-day S&P 500 returns: the data cannot decide (CI $[-22; 41]$ bp)
- ▣ Next: frontiers, and how to read a top-journal ML result

The Virtue of Complexity

- Kelly, Malamud & Zhou (2024): predict the market return with ridge regressions on P random Fourier features (nonlinear transformations) of a few predictors
 - ▶ rolling training windows as short as $T = 12$ months, with P in the thousands
- Theory: “complex” models with more parameters than observations ($P > T$) dominate simple ones; the out-of-sample R^2 may be negative while the Sharpe ratio of the timing strategy **rises** with complexity
 - ▶ beyond the interpolation point the minimum-norm solution shrinks implicitly: “double descent”
- Empirically: more complex models give higher out-of-sample Sharpe ratios of market timing
- Challenge to the bias–variance intuition of Section 2: complexity is not automatically overfitting
- Survey of the whole field: Kelly & Xiu (2023)

Reading a Top-Journal ML Result

- Nagel (2025): for $P \gg T$ the forecast is a similarity-weighted average of the T training returns
 - ▶ in short windows similarity is mostly recency, and it falls with predictor volatility: a **volatility-timed momentum** strategy
 - ▶ on artificial data with reversals the method builds the same strategy, which then loses
- Avramov, Cheng & Metzker (2023): deep-learning signal profits come from hard-to-arbitrage stocks; excluding microcaps, distressed stocks or high-volatility episodes attenuates them, and trading costs erode them further
- Chen, Pelger & Zhu (2024): impose economic structure instead; the no-arbitrage moment conditions of the stochastic discount factor become the loss of a deep network
- Questions to ask of any result
 - ▶ which benchmark (zero, historical mean, buy-and-hold)? which training window?
 - ▶ tradable after costs? what simpler strategy does the model reduce to?
 - ▶ which test, and how many trials?

Deep Learning for Sequences

Recurrent networks (LSTM, GRU)

- ▣ Hidden state carries memory: $h_t = g(h_{t-1}, x_t)$
- ▣ Gates control what is kept or forgotten, mitigating vanishing gradients
- ▣ Used for volatility, limit-order-book and intraday forecasting

Transformers

- ▣ Self-attention: $\text{softmax}(QK^\top / \sqrt{d_k})V$, with query, key, value matrices Q, K, V
- ▣ Long-range dependence, parallel training
- ▣ Backbone of large language models (Chapter 15)

Time-Series Foundation Models

Pre-trained forecasters

- ▣ Pre-trained on large collections of series, used **zero-shot** (without re-training): Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
 - ▣ Useful baselines for volatility and volume forecasting
 - ▣ For returns they inherit the same signal-to-noise limits
 - ▣ Pre-training data may overlap your test period: a new form of leakage
-
- ▣ Covered in depth, with the risk-forecast evaluation tools of Chapter 8, in Chapter 14

Reinforcement Learning, Generative Models, XAI, Risk

Reinforcement learning (RL)

- ▣ A policy $\pi(a | s)$ chooses actions a in states s to maximise cumulative reward
- ▣ Optimal execution, market making, deep hedging: Buehler et al. (2019)

Generative models

- ▣ GANs, variational autoencoders (VAEs), diffusion models: synthetic scenarios for stress tests

Explainable AI (XAI)

- ▣ Required by regulators (EU AI Act, credit scoring, model risk)
- ▣ SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

ML for risk

- ▣ Value-at-Risk (VaR): conditional quantiles (quantile forests, quantile-loss boosting); Expected Shortfall (ES): average of fitted tail quantiles or a joint VaR–ES loss; volatility: a squared-return target (Chapters 7–8)

▣ Quantinar:  Reinforcement Learning  GANs  Explainable AI  ML in Financial Risk

Deep Learning in Asset Pricing: The Research Question

- **Chen, Pelger & Zhu (2024)**: estimate the stochastic discount factor (SDF) M_{t+1} that prices **all individual US stocks**, conditioning on many firm characteristics and on the whole history of macroeconomic series (Introduction)
 - ▶ No arbitrage: $E_t[M_{t+1}R_{t+1,i}^e] = 0$ for every excess return $R_{t+1,i}^e$ (Section I.A)
 - ▶ With $M_{t+1} = 1 - \omega_t^\top R_{t+1}^e$, the weights ω_t are those of the conditional tangency portfolio $F_{t+1} = \omega_t^\top R_{t+1}^e$ (eq. 1), so $E_t[R_{t+1,i}^e] = \beta_{t,i} E_t[F_{t+1}]$
- Builds on **Gu, Kelly & Xiu (2020)**: their best network forecasts $E_t[R_{t+1,i}^e]$ by least squares and is the benchmark here, the feedforward network (FFN)
 - ▶ Forecasting fits variation (a second moment) where the signal-to-noise ratio is low; the no-arbitrage condition targets the risk premium (a first moment), and averaging over time removes much of the noise (Introduction)
- Why it matters: one estimator for three open questions of asset pricing
 - ▶ the functional form of the SDF; the right test assets; the states of the economy
- Links to this chapter
 - ▶ Networks, dropout and ensembles (Machine Learning Toolbox); a time-ordered split instead of shuffled folds
 - ▶ The lesson of “Reading a Top-Journal ML Result”: economic structure before flexibility

Deep Learning in Asset Pricing: Data and Method

Data (Section III.A)

- Monthly returns of all CRSP stocks, January 1967 – December 2016; excess over the one-month Treasury-bill rate
- 46 firm characteristics in six groups: past returns, investment, profitability, intangibles, value, trading frictions (Table A.II)
 - ▶ Only stock-months with all 46 available: about 10,000 of 31,000 stocks; ranked each month into cross-sectional quantiles
- 178 macroeconomic series: 124 of FRED-MD (McCracken & Ng (2016)), the 46 monthly medians of the characteristics, 8 predictors of Welch & Goyal (2008); transformed to stationarity
- Split: training 1967–1986, validation 1987–1991, test 1992–2016

Method (Sections I–II, Appendix C)

- Adversarial generalized method of moments: $\min_{\omega} \max_g \frac{1}{N} \sum_j (E[M_{t+1} R_{t+1,j}^e g(l_t, l_{t,j})])^2$ (eq. 3); g builds the test assets with the largest pricing errors, motivated by Hansen & Jagannathan (1997)
- Networks: an FFN for ω ; an LSTM that turns the macro history into 4 hidden states; the adversary g with 8 instruments; three steps: unconditional SDF, maximise over g , minimise again; the full model is labelled GAN (Section II.D)
- Tuning: 384 configurations ranked by the validation Sharpe ratio (SR); chosen: 2 layers of 64 units, dropout, Adam; ensemble of 9 fits (Table A.VIII); loadings: a second FFN fits $E_t[R_{t+1,i}^e F_{t+1}] \propto \beta_{t,i}$
- Benchmarks: the FFN of Gu, Kelly & Xiu (2020); the linear SDF $\omega_{t,i} = \theta^\top l_{t,i}$ on long and short characteristic legs, unregularised (LS) or elastic net (EN); FF3/FF5 tangency portfolios (Fama & French (2015))
- Metrics: SR of F ; $EV = 1 - \overline{\hat{\epsilon}^2} / (\overline{R^e})^2$ (time averages of monthly cross-sectional means); cross-sectional R^2 : one minus mean squared time-averaged residual over mean squared time-averaged excess return, stocks weighted by T_i / T (II.F)

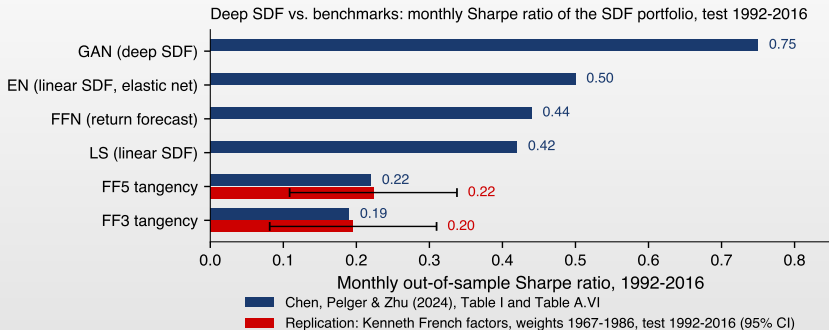
Deep Learning in Asset Pricing: Key Results, 1992–2016

- Source: the latest public version, [arXiv:1904.00745v6](https://arxiv.org/abs/1904.00745v6) (August 2021); tables and figures numbered as there
- Monthly out-of-sample SR of the SDF (Table I): GAN 0.75, EN 0.50, FFN 0.44, LS 0.42; FF5 0.22, FF3 0.19 (Table A.VI)
 - ▶ Reading: annualised 2.6 vs. 1.7 (EN) and 1.5 (FFN); the regularised linear no-arbitrage model (EN) already beats the forecasting network; unregularised LS does not
- Individual stocks (Table I): EV 8% (GAN) vs. 3–4%; cross-sectional R^2 23% vs. 14–19%
 - ▶ Reading: the no-arbitrage objective improves both fits, although it maximises neither
- Deciles sorted on the loadings β (Table II, test period, annualised): top minus bottom 0.39; FF5 alpha 0.39, $t = 9.96$; the GRS test rejects the CAPM, FF3 and FF5 ($p = 0.00$)
 - ▶ Reading: loadings predict returns, linearly ($R^2 = 0.97$ of mean return on β , Figure 8), and the factor models do not price them
- 46 sets of characteristic-sorted deciles (Table III): GAN cross-sectional R^2 above 90% for all; EV short-term reversal 0.70 vs. 0.43 (EN), momentum 0.54 vs. 0.26
 - ▶ Reading: gains come from extreme deciles and interactions; size and book-to-market deciles are easy for every model (Section III.E)
- Macro states (Figure 6): the 178 raw series make the out-of-sample SR collapse; GAN loses about 10% without macro states and 20% with unconditional moments only
 - ▶ Reading: the dynamics of macro data must be modelled (LSTM), not fed as the last increment
- Largest stocks (Table IV, monthly SR, models fitted on all stocks, evaluated on the subset): about 1,500 stocks GAN 0.41, FFN 0.30, EN 0.15; about 550 stocks 0.26, 0.24, 0.23; re-estimated on the 1,500: 0.42, 0.28, 0.42
 - ▶ Reading: the edge narrows sharply among the largest stocks; EN catches up when re-estimated on them

Deep Learning in Asset Pricing: Replication, Limits, Extensions

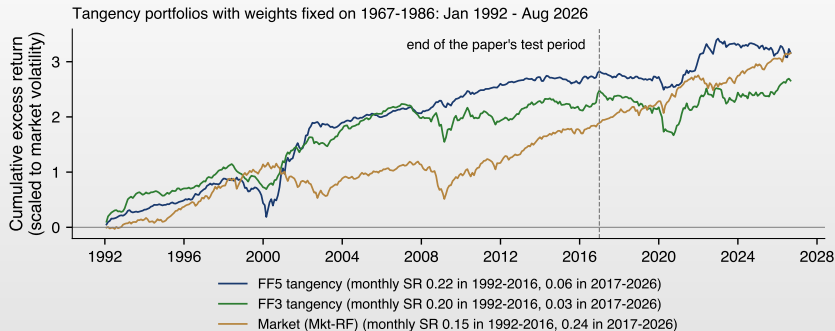
- Data limitation: the stock-level panel (CRSP returns, 46 characteristics) is not in the course data, so the GAN on individual stocks cannot be reproduced
- With the [Kenneth French Data Library](#) and FRED-MD, on the published split
 - ▶ Replicable: FF3 and FF5 tangency weights estimated on 1967–1986 (footnote 23), monthly SR on 1992–2016 next to Table A.VI
 - ▶ Not replicable with the course data: the published LS and EN (Tables I and III) use rank-weighted legs of 46 characteristics and stock-level betas
 - ▶ Analogue on portfolio data, an extension rather than a replication: the linear SDF ($\hat{\theta}_{LS}$ in closed form, $\hat{\theta}_{EN}$ with the objective of Section II.F) on halves (deciles 1–5, 6–10) of the 15 characteristics that the library sorts into deciles; its levels are not comparable with Tables I and III
- Limitations stated by the authors
 - ▶ The positivity of the SDF is not imposed (footnote 9); only stock-months with all characteristics, which mainly drops very small stocks (Section III.A)
 - ▶ Frictions (Section III.I, Figure 17): annual SR 1.73 without the smallest 40% of stocks, 2.07 without the widest 40% bid–ask spreads, 1.87 without the lowest 40% turnover; see also [Avramov, Cheng & Metzker \(2023\)](#)
 - ▶ In-sample metrics exceed the out-of-sample ones, a sign of some overfitting (Section III.C); the hidden states are not comparable across fits, so they cannot be averaged (Section II.E)
- Open extension: freeze the 1967–1991 design and evaluate 2017–2026, after the paper's sample; does the linear no-arbitrage SDF still beat FF5, and do LSTM states of FRED-MD add to it?

Deep Learning in Asset Pricing: Sharpe Ratios, Paper and Replication



- Blue: the paper (Table I, Table A.VI); red: FF3 and FF5 tangency portfolios $w \propto \hat{\Sigma}^{-1} \hat{\mu}$ on the [Kenneth French](#) factors, weights fixed on 1967–1986, monthly SR on 1992–2016 ($T = 300$)
- Ours: FF5 0.22, 95% CI [0.11; 0.34]; FF3 0.20, CI [0.08; 0.31] (SE 0.058, [Lo \(2002\)](#)); paper: 0.22 and 0.19

Deep Learning in Asset Pricing: The Fama–French Benchmarks After 2016



- Same fixed weights, January 1992 – August 2026; each series scaled to the volatility of the market excess return (Mkt-RF) on 1992–2016, cumulative sum of monthly excess returns
- Monthly SR, 2017–2026 ($T = 116$, $SE \approx 0.09$): FF5 0.06, FF3 0.03, market 0.24; on 1992–2016: 0.22, 0.20, 0.15

Deep Learning in Asset Pricing: Reading the Two Charts

- The Fama–French benchmarks of the paper are reproduced on public data: both published values lie inside the 95% CIs of the replication
 - ▶ GAN 0.75 vs. FF5 0.22: with Lo SEs 0.065 and 0.058, the SE of the difference is at most 0.12 whatever their correlation, so the gap exceeds 4 SEs (i.i.d. Normal returns); a paired test on aligned returns (Ledoit & Wolf (2008)) is sharper
- The fixed FF5 weights load mainly on profitability and investment: RMW 0.42, CMA 0.30 (weights scaled to $\sum |w| = 1$)
 - ▶ FF3 puts 0.61 on HML; both combinations were chosen on 1967–1986
- After 2016 the fixed tangency portfolios have monthly Sharpe ratios far below the market's
 - ▶ With $T = 116$ months the SE is about 0.09: a warning, not yet a test
 - ▶ The question for the GAN is the same: does a design frozen in 1991 survive out of sample, after the paper?

An Open Question: Discovery or Data Mining at Scale?

- [Chen, Lopez-Lira & Zimmermann \(2022\)](#): mining 29,000 accounting ratios for $t > 2$ gives cross-sectional predictability similar to peer-reviewed predictors; for both, about half of it remains after the original samples
- [Novy-Marx & Velikov \(2025\)](#): an LLM pipeline mines over 30,000 signals, keeps 96 that pass a fixed protocol and writes hundreds of complete papers, with invented motivations and some imagined citations
 - ▶ HARKing (hypothesising after the results are known) at industrial scale
- Open: when a machine proposes the features, the model and the story, what is the number of trials N , and what evidence still separates a discovery from the best of many draws (DSR and PBO, Backtest Overfitting)?
- Formally: $H_{0,j} : \beta_j = 0, j = 1, \dots, M$; control the FWER (Holm) or the FDR ([Benjamini & Hochberg \(1995\)](#)); [Harvey, Liu & Zhu \(2016\)](#) require a new factor to have $t > 3$

The Discovery Loop, Step by Step (1/2)

- **Literature:** "Summarise the evidence on how much published return predictability survives out of sample and after publication: DOI or arXiv id, sample, headline decay estimate."
 - ▶ Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search
- **Hypotheses:** "For a feature that predicts S&P 500 returns in 2001-2015, give competing explanations (risk premium, reversal, microstructure, data mining) and what each predicts for 2016-2026 and for other indices."
- **Data and code:** "Write Python that builds 34 technical features from data/market/GSPC.INDX.csv and VIX.INDX.csv with information up to day t only, and reports Newey-West t-statistics for 1-, 5- and 20-day returns in 2001-2015."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); first reproduce the case study: "always up" 58.1%

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Write the pre-analysis plan: the full list of features and horizons, the holdout, the corrections (Holm, BH, max-t) and the result that would count as a failure."
 - ▶ Placebo: shifting the target circularly (250–3,000 days) breaks its alignment with the features but keeps each series' own dependence; a valid null for $\max_j |t_j|$ if the target is stationary and the shifts exceed its dependence range
- **Critique**: "Act as a referee: which of these significant features are one effect counted several times, and which are known anomalies rather than new ones?"
- **Report**: all M tests, all trials including the discarded ones, DSR and PBO for any strategy built on them

What the Researcher Must Check

- ▣ **Leakage:** features standardised or selected on the full sample
 - ▶ Test: fit every transformation inside the training fold; purged CV with embargo (Cross-Validation in Finance)
- ▣ **Undercounted trials:** the assistant silently drops failed variants
 - ▶ Test: log every proposal; M in the correction counts them all
- ▣ **Correlated tests:** twenty windows of one effect look like twenty discoveries
 - ▶ Test: $\max_j |t_j|$ under a joint null, or cluster the features first
- ▣ **Old wine, new name:** a “new” feature is a known anomaly, or its reference does not exist
 - ▶ Test: resolve every DOI; compare with the published anomaly lists
- ▣ **Label overlap:** labels spanning several days leak into the validation fold
 - ▶ Test: purging; the leakage experiment of this chapter (69% on pure noise)

Mini-Case: 102 Tests, One Old Survivor

- A typical assistant proposal: “add more indicators: momentum, moving-average distances, volatility ratios, RSI and VIX changes over several windows”
 - ▶ We test 34 features $\times$ 3 horizons (1, 5, 20 days) = 102 predictive regressions on 2001–2015, Newey–West t
- $|t| > 1.96$: **21** of 102, against 5.1 expected under the global null at 5%
- Holm (FWER 5%): **0**; BH (FDR 10%): **13**; $|t| > 3$: **4**
- Joint null (500 circular shifts of the target): 95% quantile of $\max_j |t_j| = 3.32$; one survivor, the 1-day return on the next day's return, $t = -3.37$: short-term reversal, a known effect
- Holdout 2016–2026: 3 of the 21 keep their sign with $|t| > 1.96$, all reversal variants ($t = -2.17$ for the 1-day return); the VIX-based “finds” fail
- Reading: Holm (FWER 5%) and BH (FDR 10%) control different error rates at different levels, so they need not agree; BH's guarantee needs independence or positive regression dependence of the true-null p-values, which correlated predictors do not ensure. The only robust result is an old one

 `ai_discovery_case.py`

Your Turn: a Project Seed

- Question: can an AI-assisted feature search find a predictor of index returns (S&P 500, BET, Bitcoin) that survives a pre-registered correction and a holdout, beyond short-term reversal?
- Replicate first: the case study of this chapter ("always up" 58.1%, Logit 57.6%, AUC 0.478) and this mini-case (21 of 102, one survivor)
- Deliverables
 - ▶ A feature list fixed before the data are touched; all tests with Holm, BH and the max- $|t|$ null; a holdout
 - ▶ DSR and PBO for any strategy built on the survivors; a referee-style critique and your reply
- Log each prompt that shaped a decision in AI_USE.md; each invented reference, leak or undercounted trial caught in AI_ERRORS.md, with the test that caught it ([templates](#))

Best-Practice Checklist

Data and features

- ☐ Point-in-time data, no survivorship bias
- ☐ Memory estimated with a CI (local Whittle), not an ADF rule
- ☐ Transformations fitted on training folds only

Labels

- ☐ Volatility-scaled, path-dependent labels
- ☐ Record label spans; weight by uniqueness

Validation

- ☐ Purged K-fold / CPCV with embargo
- ☐ Always compare with naive baselines
- ☐ AUC and log-loss with block-bootstrap CIs
- ☐ Test against the benchmark: PT, DM / CW / GW

Backtesting

- ☐ Include transaction costs and slippage
- ☐ Log every trial; report N
- ☐ DSR, SPA and PBO before any claim

Conclusions

- ▣ ML offers flexible models, but financial data offer **little signal and much noise**
- ▣ Classical tools (LASSO, random forests, boosting, neural nets) remain the workhorses; the finance-specific part is the **process**
- ▣ An ADF rejection is not stationarity: truncated FFD of the S&P 500 log price stays $I(1)$; estimate memory with local Whittle
- ▣ Triple-barrier and meta-labelling produce labels that reflect real trading
- ▣ Shuffled K-fold can report 69% accuracy on pure noise: use purging and embargo
- ▣ On the S&P 500 no model beats “always up” (PT, DM, GW); the best of many backtests must be deflated or bootstrapped (DSR, SPA)

Next

- ▣ Chapter 14: deep learning and time-series foundation models; Chapter 15: large language models and sentiment

Further Study on Quantinar

Advanced courses on quantinar.com

-  Random Forests
-  Shapley Values
-  Shapley Curves: A Smoothing Perspective
-  Machine Learning in Financial Risk
-  Introduction to Reinforcement Learning
-  Generative Adversarial Networks
-  MSCA Digital: The Need for Explainable AI
-  Business Analytics and Data Science
-  Model Selection Criteria
-  PAC Learning

References (1/9)

- ▣ Ané, T., Geman, H. (2000). Order flow, transaction clock, and normality of asset returns. *Journal of Finance*, 55(5), 2259–2284.
- ▣ Ansari, A.F. et al. (2024). Chronos: learning the language of time series. *arXiv:2403.07815*.
- ▣ Avramov, D., Cheng, S., Metzker, L. (2023). Machine learning vs. economic restrictions: evidence from stock return predictability. *Management Science*, 69(5), 2587–2619.
- ▣ Bailey, D.H., Borwein, J., López de Prado, M., Zhu, Q.J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4), 39–69.
- ▣ Bailey, D.H., López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Będowska-Sójka, B., Wójcik, P., Pele, D.T. (2026). Early warning systems for cryptocurrency markets: predicting ‘zombie’ assets using machine learning. *North American Journal of Economics and Finance*, 81, 102543.

References (2/9)

- ▣ Belloni, A., Chernozhukov, V., Hansen, C. (2014). [Inference on treatment effects after selection among high-dimensional controls](#). *Review of Economic Studies*, 81(2), 608–650.
- ▣ Benjamini, Y., Hochberg, Y. (1995). [Controlling the false discovery rate: a practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Bergmeir, C., Hyndman, R.J., Koo, B. (2018). [A note on the validity of cross-validation for evaluating autoregressive time series prediction](#). *Computational Statistics & Data Analysis*, 120, 70–83.
- ▣ Breiman, L. (2001). [Random forests](#). *Machine Learning*, 45(1), 5–32.
- ▣ Buehler, H., Gonon, L., Teichmann, J., Wood, B. (2019). [Deep hedging](#). *Quantitative Finance*, 19(8), 1271–1291.
- ▣ Campbell, J.Y., Thompson, S.B. (2008). [Predicting excess stock returns out of sample: can anything beat the historical average?](#) *Review of Financial Studies*, 21(4), 1509–1531.
- ▣ Campbell, M., Hoane, A.J., Hsu, F. (2002). [Deep Blue](#). *Artificial Intelligence*, 134(1–2), 57–83.

References (3/9)

- Chen, A.Y., Lopez-Lira, A., Zimmermann, T. (2022). Does peer-reviewed research help predict stock returns? *arXiv:2212.10317*.
- Chen, L., Pelger, M., Zhu, J. (2024). Deep learning in asset pricing. *Management Science*, 70(2), 714–750.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal*, 21(1), C1–C68.
- Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- Clark, T.E., West, K.D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311.
- Das, A., Kong, W., Sen, R., Zhou, Y. (2024). A decoder-only foundation model for time-series forecasting. *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235.
- DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44(3), 837–845.

References (4/9)

- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Dixon, M.F., Halperin, I., Bilokon, P. (2020). *Machine Learning in Finance: From Theory to Practice*. Springer.
- ▣ Fama, E.F., French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- ▣ Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Friedman, J.H. (2001). Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.

References (5/9)

- Hansen, L.P., Jagannathan, R. (1997). *Assessing specification errors in stochastic discount factor models*. *Journal of Finance*, 52(2), 557–590.
- Hansen, P.R. (2005). *A test for superior predictive ability*. *Journal of Business & Economic Statistics*, 23(4), 365–380.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). *...and the cross-section of expected returns*. *Review of Financial Studies*, 29(1), 5–68.
- Hastie, T., Tibshirani, R., Friedman, J. (2009). *The Elements of Statistical Learning*. 2nd ed. Springer.
- Hoerl, A.E., Kennard, R.W. (1970). *Ridge regression: biased estimation for nonorthogonal problems*. *Technometrics*, 12(1), 55–67.
- Hooker, G., Mentch, L., Zhou, S. (2021). *Unrestricted permutation forces extrapolation: variable importance requires at least one more model, or there is no free variable importance*. *Statistics and Computing*, 31(6), 82.
- Hosking, J.R.M. (1981). *Fractional differencing*. *Biometrika*, 68(1), 165–176.
- Kelly, B., Malamud, S., Zhou, K. (2024). *The virtue of complexity in return prediction*. *Journal of Finance*, 79(1), 459–503.

References (6/9)

- Kelly, B., Xiu, D. (2023). *Financial machine learning*. *Foundations and Trends in Finance*, 13(3–4), 205–363.
- Künsch, H.R. (1989). *The jackknife and the bootstrap for general stationary observations*. *Annals of Statistics*, 17(3), 1217–1241.
- Ledoit, O., Wolf, M. (2008). *Robust performance hypothesis testing with the Sharpe ratio*. *Journal of Empirical Finance*, 15(5), 850–859.
- Lo, A.W. (2002). *The statistics of Sharpe ratios*. *Financial Analysts Journal*, 58(4), 36–52.
- Lo, A.W. (2004). *The adaptive markets hypothesis*. *Journal of Portfolio Management*, 30(5), 15–29.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- López de Prado, M. (2020). *Machine Learning for Asset Managers*. Cambridge University Press.

References (7/9)

- ▣ Lundberg, S.M., Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
- ▣ McCracken, M.W., Ng, S. (2016). FRED-MD: a monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4), 574–589.
- ▣ Nagel, S. (2025). Seemingly virtuous complexity in return prediction. NBER Working Paper 34104.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Novy-Marx, R., Velikov, M. (2025). AI-powered (finance) scholarship. NBER Working Paper 33363.
- ▣ Opdyke, J.D. (2007). Comparing Sharpe ratios: so where are the p-values? *Journal of Asset Management*, 8(5), 308–336.
- ▣ Pesaran, M.H., Timmermann, A. (1992). A simple nonparametric test of predictive performance. *Journal of Business & Economic Statistics*, 10(4), 461–465.
- ▣ Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.

References (8/9)

- ▣ Rasul, K. et al. (2023). Lag-Llama: towards foundation models for probabilistic time series forecasting. *arXiv:2310.08278*.
- ▣ Robinson, P.M. (1995). Gaussian semiparametric estimation of long range dependence. *Annals of Statistics*, 23(5), 1630–1661.
- ▣ Romano, J.P., Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4), 1237–1282.
- ▣ Rosenblatt, F. (1958). The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6), 386–408.
- ▣ Shimotsu, K. (2010). Exact local Whittle estimation of fractional integration with unknown mean and time trend. *Econometric Theory*, 26(2), 501–540.
- ▣ Shimotsu, K., Phillips, P.C.B. (2005). Exact local Whittle estimation of fractional integration. *Annals of Statistics*, 33(4), 1890–1933.
- ▣ Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society B*, 58(1), 267–288.

References (9/9)

- ▣ Wager, S., Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523), 1228–1242.
- ▣ Welch, I., Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.
- ▣ White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126.
- ▣ Woo, G. et al. (2024). Unified training of universal time series forecasting transformers. *arXiv:2402.02592*.
- ▣ Zou, H., Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society B*, 67(2), 301–320.
- ▣ **Code and data:** github.com/danpele/MFM/tree/main/Quantlets/Ch_13